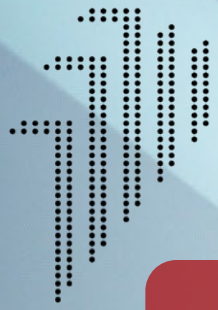
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UNIVERSITY OF APPLIED SCIENCES

Complex Event Processing (CEP)

Introduction to Artificial Intelligence

Prof. Dr. rer. nat. Hamed Shariat Yazdi

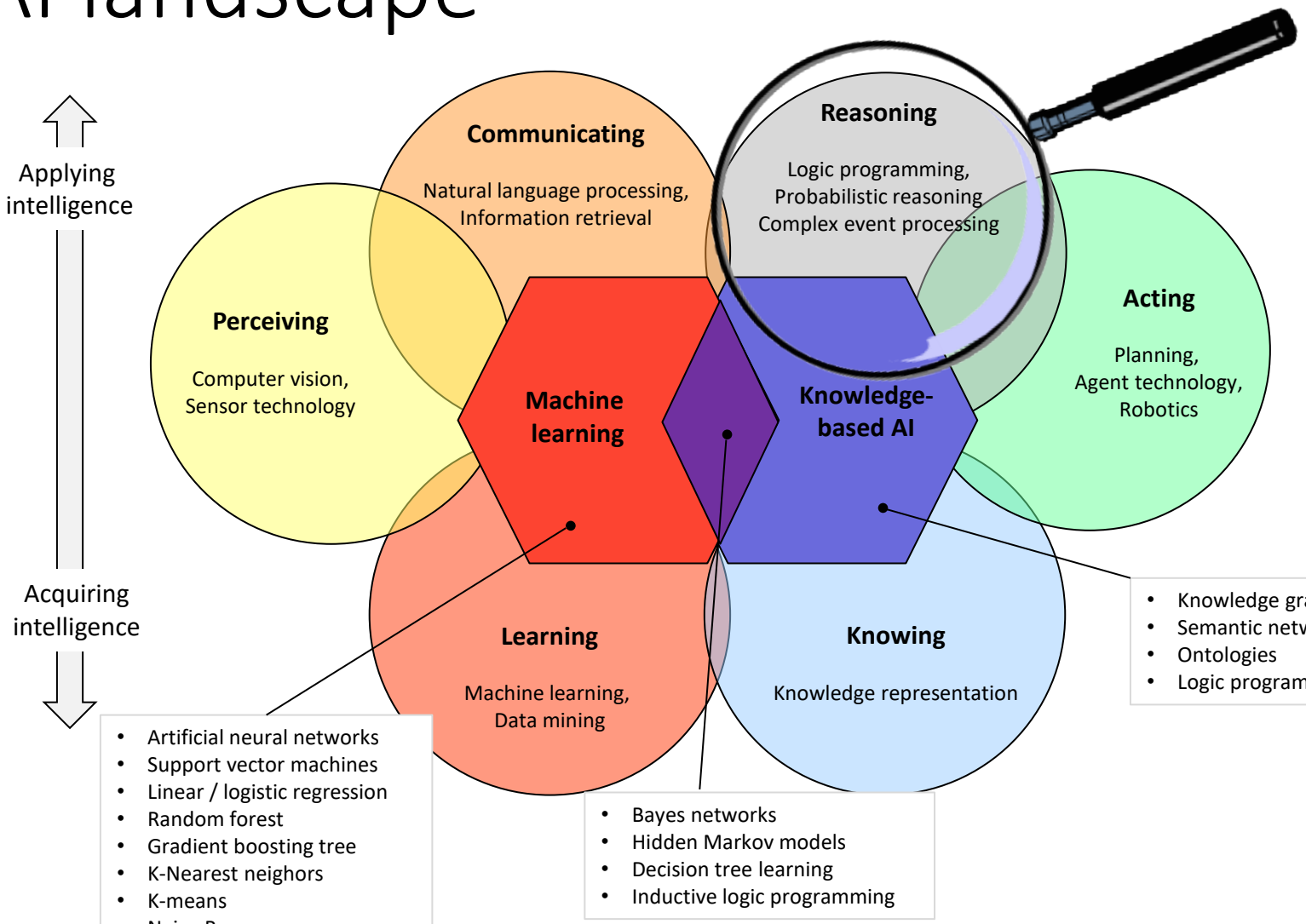
Department of Computer Science

Darmstadt University of Applied Sciences

Agenda

- Overview
- CEP applications
- CEP concepts and technologies
- Example
- Quick check

AI landscape



Inspired by https://www.sigs-datacom.de/order/poster/Landkarte-KI-Poster-AI-2018.php

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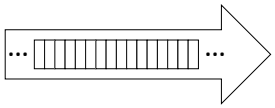
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Fraud detection



Stream of credit card transactions



CEP



Check for unusually high transactions in short time



Fraud alert: Human intervention required

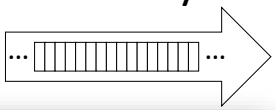


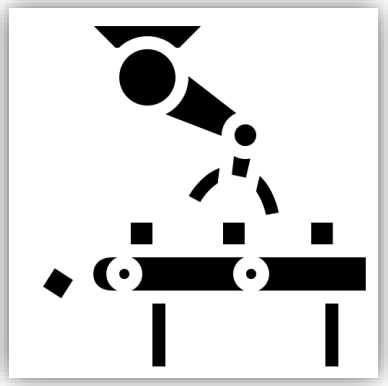


https://p2.piqsels.com/preview/377/158/322/credit-credit-card-finance-credit-cards.jpg

Predictive Maintenance

Stream of sensor inputs from machinery in factory





CEP

Check for patterns that indicate deterioration, e.g. vibrations



Warning: Part should be replaced soon



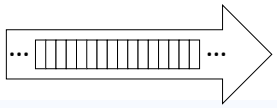
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https://game-icons.net/icons/000000/ffffff/1x1/delapouite/factory-arm.png
https://images-wixmp-ed30a86b8c4ca887773594c2.wixmp.com/

Logistics



Stream of RFID signals





CEP



Check for conditions that require certain actions

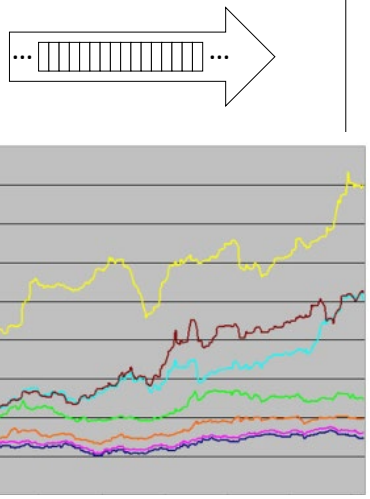
- Inform logistic partners on transport
- Inform customer on delivery
- Alert, e.g., interrupted freezer chain
- ...

https://1.bp.blogspot.com/-j2rezbiNZnI/XX0qT_mMqXI/AAAAAAAAL6E/yt764xK4mLM4xFGe0_ayJgY4FJI9T3igQCLcBGAsYHQ/s1600/Logistics%2B%2528II%2529.jpg

Stock market trading



Stream of stock market data



Classify curent situation for selling/buying actions (ML)

Sell stocks

Buy stocks



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https://www.wikimedia.de/
https://cdn.pixabay.com/photo/2015/05/31/06/53/euro-790905_960_720.jpg

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Definition of CEP

Complex Event Processing (CEP):
Processing streams of event data and deriving conclusions from them

Source: https://www.thecepblog.com/2007/05/14/what-is-complex-event-processing-part-2/

What is an event?

Event:
Something notable that happens

- Examples:
  - A financial transaction
  - An airplane landing
  - A sensor outputs a reading
  - A change of state in a database, a finite state machine
  - A key stroke
  - A historic event, e.g., the French Revolution

Source: https://www.thecepblog.com/2007/05/14/what-is-complex-event-processing-part-2/

Event object

Event object:
A data record that represents an event

- Examples:
  - A Purchase order record
  - A stock tick message
  - A record with an RFID sensor reading

- Often the term “event“ is simply used for an event object

Source: https://www.thecepblog.com/2007/05/14/what-is-complex-event-processing-part-2/

Event type (a.k.a. event class)

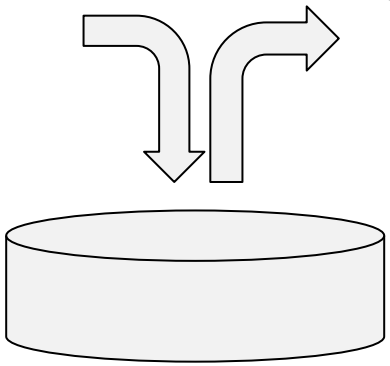
Event type (a.k.a. **event class**)
specifies the common structure of related event objects, i.e., their attributes and data types

- Example: class PurchasingEvent with following attributes
  - Timestamp
  - Buyer
  - Product
  - Price
- Often the term “event“ is simply used for event types

DBMS versus CEP

DBMS

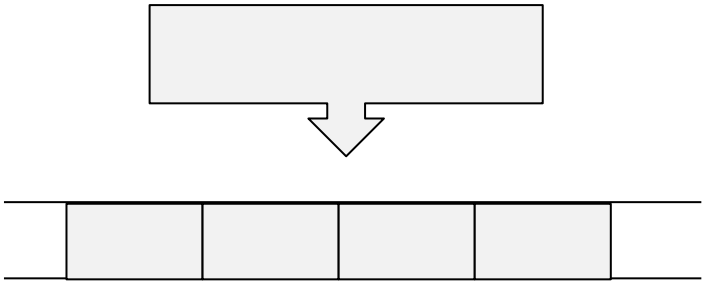
DBMS query execution: volatile



DBMS Data: persistent

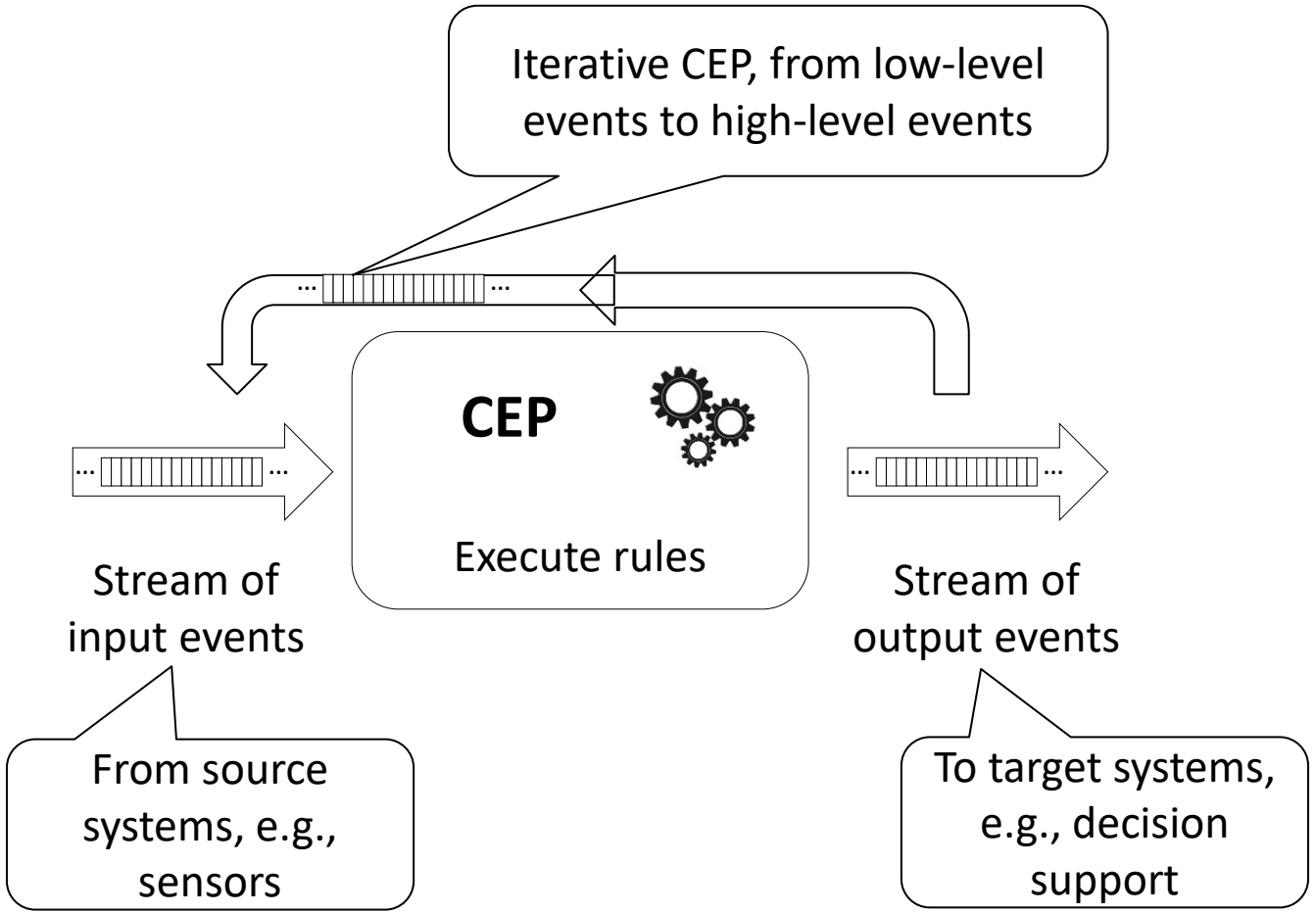
CEP

CEP rule: persistent

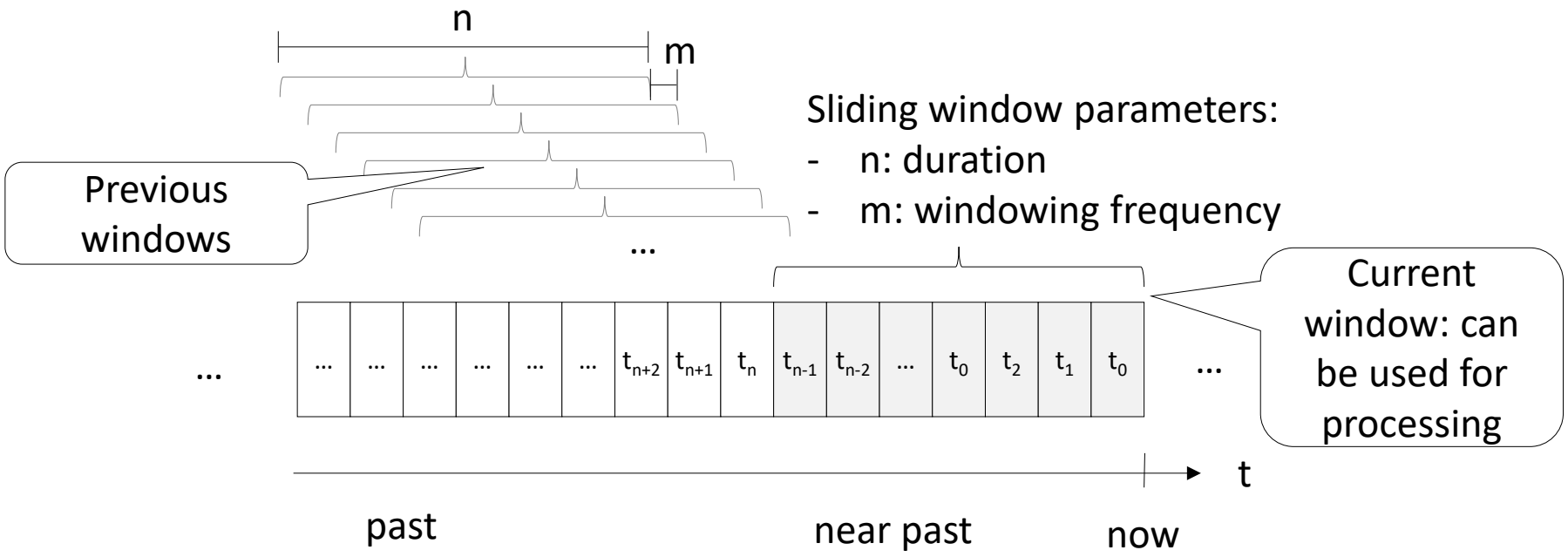


Streaming Data: flowing

Complex Event Processing



Sliding (“rolling“) window



CEP Technology

- Apache Kafka
- Apache Flink
- Apache Spark
- Drools Fusion
- MS Azure Stream Analytics
- Oracle Stream Analytics
- SAG Apama
- SAP ESP
- SAS ESP
- TIBCO
- IBM WebSphere Business Events























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CEP technology is complex

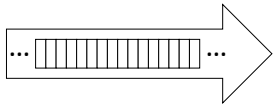
- Complexity due to inherent distribution
- Difficult to install and to operate
- → IAI lab: Use of Python library to practice sliding windows instead of using full-fledged CEP technology

Agenda

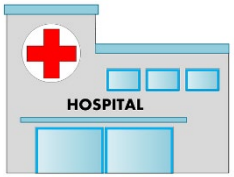
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Covid-19 Monitoring

Stream of notifications from test centers and hospitals







CEP

Compute incidence and criticality

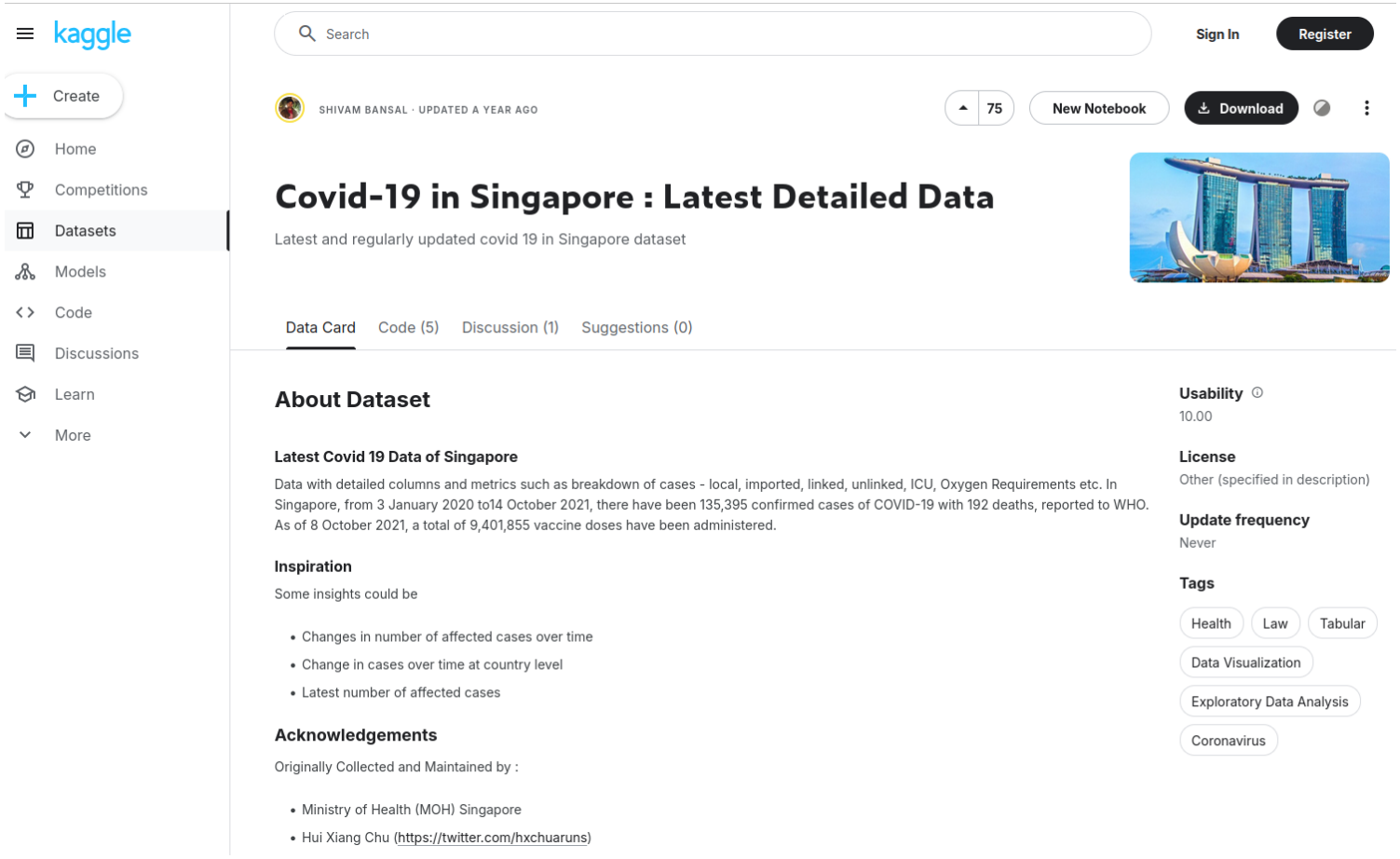
Governmental measures



https://1.bp.blogspot.com/-j2rezbiNZnI/XX0qT_mMqXI/AAAAAAAAL6E/yt764xK4mLM4xFGe0_ayJgY4FJI9T3igQCLcBGAsYHQ/s1600/Logistics%2B%2528III%2529.jpg

Dataset: COVID-19 Singapore

https://www.kaggle.com/datasets/shivamb/covid19-in-singapore-latest-detailed-data



Current data since Covid-19 break-out
Numerous features

Incidence

2020-01 – 2022-12

Hospitalization

Death rate

More features

Date	Daily Confirmed	False Positives Found	Cumulative Confirmed	Daily Discharged	Passed but not due to COVID	Cumulative Discharged	Discharged to Isolation	Still Hospitalised	Daily Deaths	Cumulative Deaths	Tested positive demise	Daily Imported	Daily Local transmission	Local cases residing in dorms MOH report	Local cases not residing in doms MOH report	Intensive Care Unit (ICU)
2020-10-11	10	0	57876	7	0	57727	100	14		27	0	7	3	3	0	1
2020-10-12	4	0	57880	23	0	57750		18		28	0	2	2	1	1	
2020-10-13	4	0	57884	12	0	57762		5		28	0	4	0	0	0	
2020-10-14	5	0	57889					9								
2020-10-15	3	0	57892					6								
2020-10-16	9	0	57901					0								
2020-10-17	3	0	57904					7								
2020-10-18	7	0	57911					10								
2020-10-19	4	0	57915	12	0	57841	34	4	0	28	0	4	0	0	0	0
2020-10-20	6	0	57921	0	0	57841	36	8	0	28	0	4	2	2	0	0
2020-10-21	12	0	57933	2	0	57843	41	13	0	28	0	12	0	0	0	0
2020-10-22	8	0	57941	8	0	57851	45	9	0	28	0	7	1	1	0	0
2020-10-23	10	0	57951	3	0	57854	39	22	0	28	0	10	0	0	0	0
2020-10-24	14	0	57965	12	0	57866	40	23	0	28	0	11	3	1	2	0
2020-10-25	5	0	57970	14	0	57880	33	21	0	28	0	3	2	2	0	0
2020-10-26	3	0	57973	21	0	57901	23	13	0	28	0	2	1	1	0	0
2020-10-27	7	0	57980	4	0	57905	26	13	0	28	0	6	1	0	1	0
2020-10-28	7	0	57987	7	0	57912	29	10	0	28	0	7	0	0	0	0
2020-10-29	7	0	57994	9	0	57921	30	7	0	28	0	7	0	0	0	0
2020-10-30	9	0	58003	10	0	57931	24	12	0	28	0	7	2	1	1	0
2020-10-31	12	0	58015	4	0	57935	32	12	0	28	0	12	0	0	0	0
2020-11-01	4	0	58019	11	0	57946	21	16	0	28	0	4	0	0	0	0
2020-11-02	1	0	58020	0	0	57946	22	16	0	28	0	1	0	0	0	0
2020-11-03	9	0	58029	13	0	57959	26	8	0	28	0	7	2	2	0	0
2020-11-04	7	0	58036	1	0	57960	26	14	0	28	0	5	2	2	0	0
2020-11-05	7	0	58043	11	0	57971	26	10	0	28	0	6	1	0	1	0
2020-11-06	4	0	58047	10	0	57981	24	6	0	28	0	4	0	0	0	0

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Loading and preparing data

```
import pandas as pd
from datetime import datetime

df = pd.read_csv('data/Covid-19_SG.csv')

df.index = df['Date'].apply(lambda s: datetime.strptime(s, '%Y-%m-%d'))
```

Convert date strings to datetime and use as DataFrame index

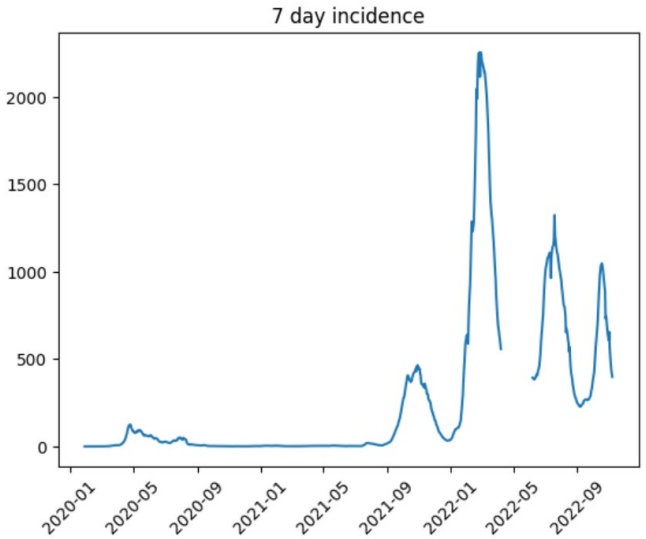
7-day-incidence

- Definition 7-day-incidence:

$$7inc = \sum_{i=1}^{7} inc_i * \frac{100{,}000}{population}$$

Sliding window of 7 days

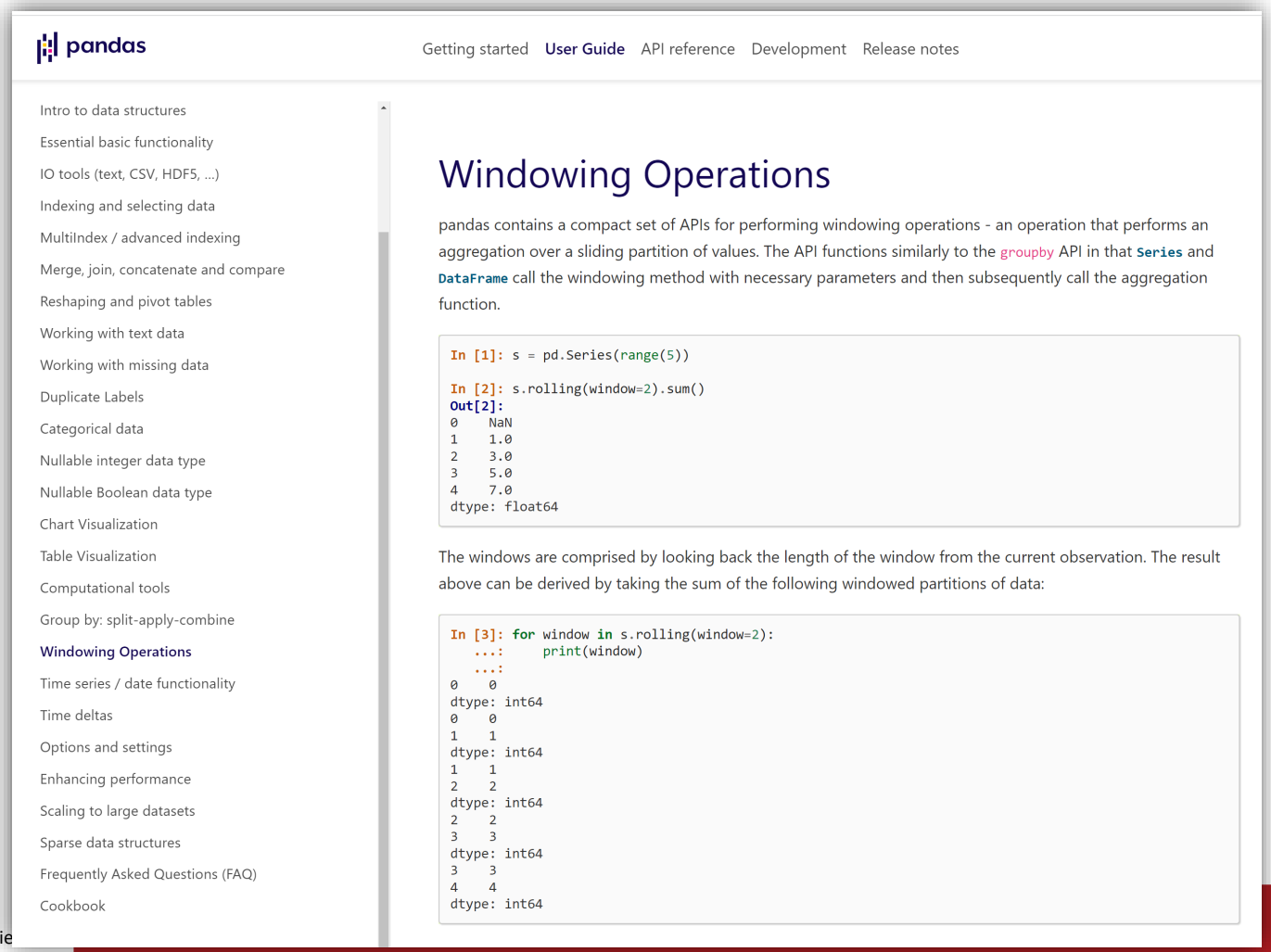
- Population Singapore: 5,686 Milion



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Pandas DataFrame windowing

https://pandas.pydata.org/pandas-docs/stable/user_guide/window.html



pandas

Getting started | User Guide | API reference | Development | Release notes

- Intro to data structures
- Essential basic functionality
- IO tools (text, CSV, HDF5, ...)
- Indexing and selecting data
- MultiIndex / advanced indexing
- Merge, join, concatenate and compare
- Reshaping and pivot tables
- Working with text data
- Working with missing data
- Duplicate Labels
- Categorical data
- Nullable integer data type
- Nullable Boolean data type
- Chart Visualization
- Table Visualization
- Computational tools
- Group by: split-apply-combine
- Windowing Operations
- Time series / date functionality
- Time deltas
- Options and settings
- Enhancing performance
- Scaling to large datasets
- Sparse data structures
- Frequently Asked Questions (FAQ)
- Cookbook

Windowing Operations

pandas contains a compact set of APIs for performing windowing operations - an operation that performs an aggregation over a sliding partition of values. The API functions similarly to the groupby API in that `Series` and `DataFrame` call the windowing method with necessary parameters and then subsequently call the aggregation function.

```
In [1]: s = pd.Series(range(5))

In [2]: s.rolling(window=2).sum()
Out[2]:
0    NaN
1    1.0
2    3.0
3    5.0
4    7.0
dtype: float64
```

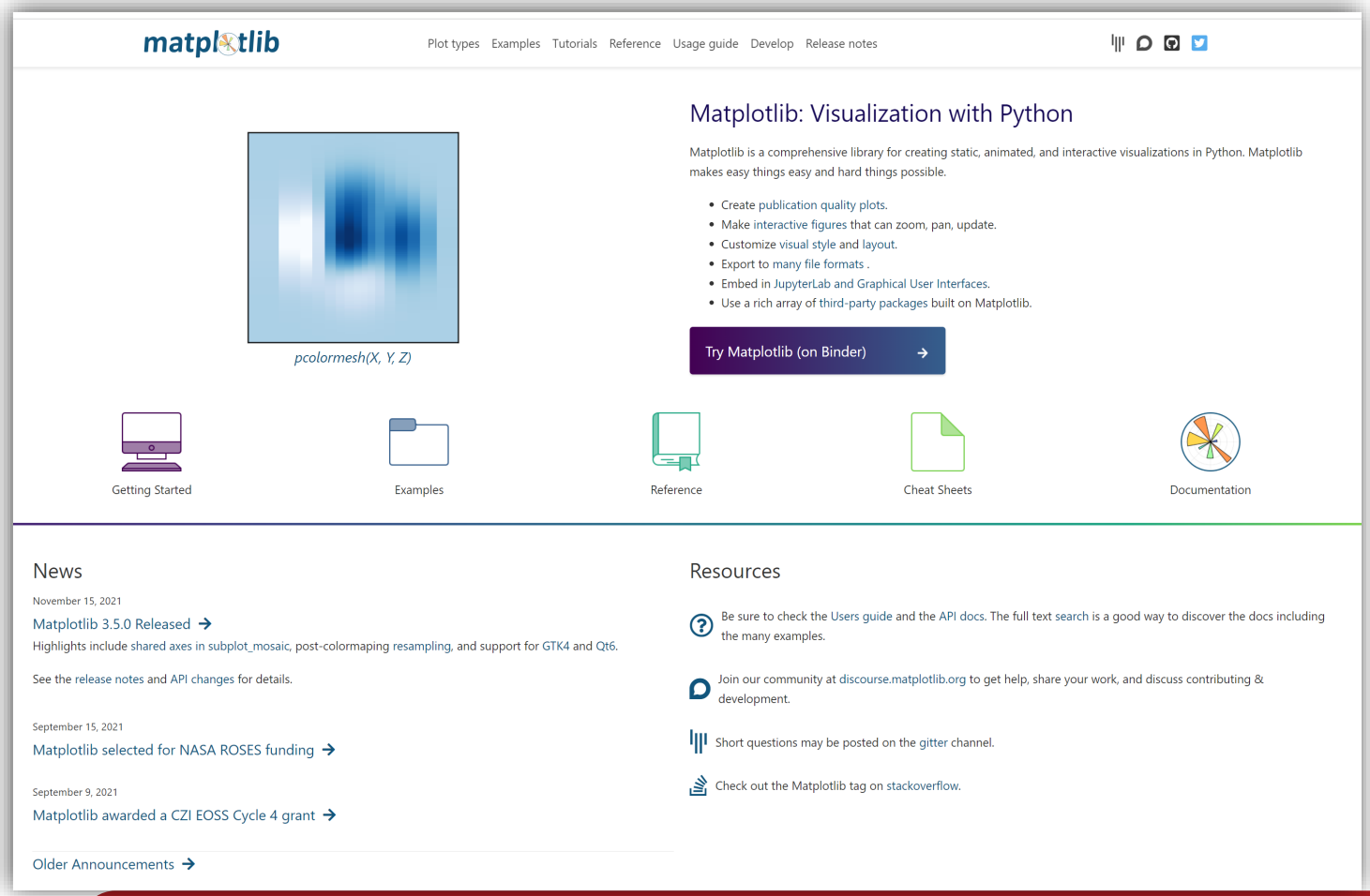
The windows are comprised by looking back the length of the window from the current observation. The result above can be derived by taking the sum of the following windowed partitions of data:

```
In [3]: for window in s.rolling(window=2):
   ...:     print(window)
   ...:
0    0
dtype: int64
0    0
1    1
dtype: int64
1    1
2    2
dtype: int64
2    2
3    3
dtype: int64
3    3
4    4
dtype: int64
```

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Matplotlib: Printing graphs

https://matplotlib.org



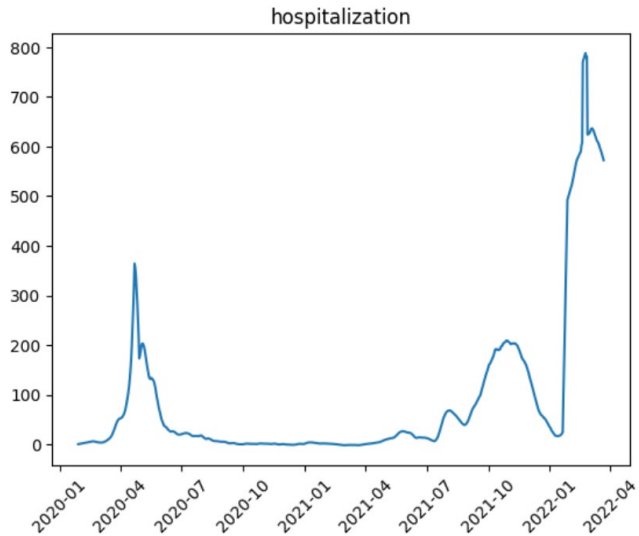
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Hospitalization rate

- Definition 7-day hospitalization rate:

$$7hosp = \sum_{i=1}^{7} hosp_i * \frac{100{,}000}{population}$$

Sliding window of 7 days



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Iterative CEP: criticality (derived from 7-day-incidence)

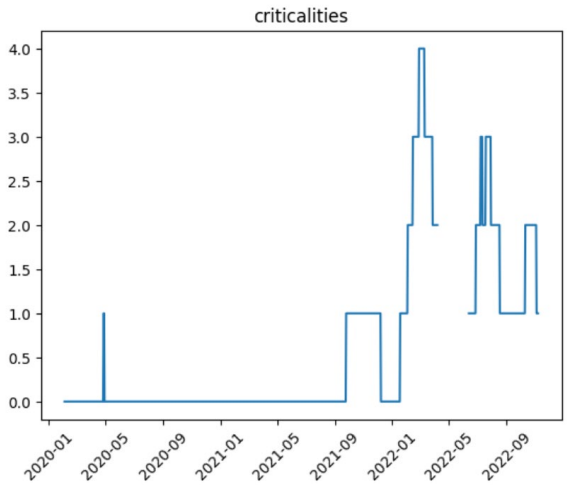
- Sample definition of criticality (to be used by government for specific measures):

$$crit(7inc) = \begin{cases} 4, if\ 7inc > 2000 \\ 3, else\ if\ 7inc > 1000 \\ 2, else\ if\ 7inc > 500 \\ 1, else\ if\ 7inc > 100 \\ 0, else \end{cases}$$

- 7-day criticality score

Sliding window of 7 days

$$7crit = min_{i=1}^{7} crit(7inc_i)$$



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Quick check
Complex Event Processing

1. What is CEP?
2. Name applications of CEP
3. What is an event, event object and event type?
4. How does iterative CEP work?
5. What are sliding (“rolling”) windows? How are they used in pandas?
6. Name CEP technologies